

Ecology flashcards, learning targets 1A–2C

How to use. Cut along the solid lines and fold each card on the dashed line so the term is on one side and the meaning on the other. Say the meaning before you flip. Sort cards into “know it” and “not yet” piles and repeat the “not yet” pile the next day. The same cards are available as an import file for Anki or Quizlet in [output/flashcards/](#).

Sets. 1A (10 cards), 1B (11 cards), 1C (13 cards), 2A (9 cards), 2B (10 cards), 2C (11 cards), Academic (26 cards). Colors match the book: blue for evidence and biodiversity, green for life and matter, orange for regulation and population.

Definitions are the same as the vocabulary pages of the study book. Terms come from the teacher’s Ecology Unit Guide.

Observation 1A	Inference 1A
Information gathered directly with the senses or with instruments. <i>"The leaf is 8 cm long."</i>	An interpretation or conclusion based on observations plus what you already know. <i>"The plant may be short because it gets little light."</i>
Data 1A	Quantitative data 1A
Recorded observations or measurements used as evidence. <i>A table of seedling heights measured each week.</i>	Data expressed as numbers, counts, or measurements. <i>12 cm; 8 seedlings; 17 °C</i>
Qualitative data 1A	Induction 1A
Data described in words or categories rather than numbers. <i>Yellow leaves; cloudy water.</i>	Reasoning from many specific observations to a broader explanation. <i>Many shaded plants grow slowly, so shade may limit growth.</i>
Deduction 1A	Annotation 1A
Reasoning from a general idea to a specific prediction that can be tested. <i>If light limits growth, plants given more light should grow taller.</i>	A note written on a text or diagram that marks a key idea, question, or connection. <i>Underlining carrying capacity and writing "changes with drought" in the margin.</i>

1A  Formative assessment	1A  Summative assessment
A low-stakes check used to guide learning and give feedback while you are still learning. <i>Homework, practice tests, and CEAs.</i>	An assessment of what you have learned at the end of a unit or course. <i>The unit exam and the final exam.</i>
1B  Organism	1B  Metabolism
An individual living thing made of one or more cells. <i>A bacterium, a rabbit, or an oak tree.</i>	All the chemical reactions in an organism that build molecules or break them down, including those that transfer usable energy. <i>Photosynthesis and cellular respiration are metabolic processes.</i>
1B  Homeostasis	1B  Stable / stability
Keeping internal conditions within a working range even while the surroundings change. <i>Body temperature stays near 37 °C in hot and cold weather.</i>	Staying within a working range over time. Stable is not the same as constant. <i>A stable body temperature still rises and falls slightly.</i>
1B  Feedback loop	1B  Negative feedback
A cycle in which a change triggers a response, and the response then affects the original condition. <i>Temperature → sweating → temperature.</i>	A response that opposes the initial change and brings the condition back toward its range. <i>Shivering when body temperature falls.</i>

Positive feedback 1B	Experimental control 1B
A response that increases the initial change until something ends the loop. <i>Contractions during childbirth strengthen until delivery.</i>	A comparison condition without the treatment, so you can see what happens on its own. <i>Bottle A with no hand in the energy-flow lab.</i>
Constant 1B	Hypothesis 1B
A factor kept the same in every condition of an experiment. <i>The same volume of water in every bottle.</i>	A testable proposed explanation. <i>"Light limits the growth of these seedlings."</i>
Prediction 1B	Ecosystem 1C
The specific result expected if the hypothesis is correct. <i>"Seedlings given more light will grow taller."</i>	All the organisms in an area plus the non-living environment and their interactions. <i>A pond: fish, plants, microbes, water, and sunlight.</i>
Biotic / abiotic 1C	Habitat 1C
Biotic: living or produced by living things. Abiotic: nonliving physical factors. <i>Biotic: frogs, dead leaves. Abiotic: temperature, water, light.</i>	The place where an organism lives. <i>The pond is the frog's habitat.</i>

Niche 1C	Autotroph / producer 1C
An organism's role: what it eats, what eats it, and the conditions and resources it uses. <i>Eating insects and being eaten by herons are parts of a frog's niche.</i>	Makes its own organic molecules from carbon dioxide using light or chemical energy. <i>Grass, algae, some bacteria.</i>
Heterotroph / consumer 1C	Decomposer 1C
Gets organic matter and energy by eating other organisms or their products. <i>Rabbit, hawk, and mushroom.</i>	A heterotroph that breaks down dead organic matter and wastes, returning nutrients to soil or water. <i>Fungi and many bacteria.</i>
Photosynthesis 1C	Cellular respiration 1C
Producers use light energy to build sugar from carbon dioxide and water and release oxygen. $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	Cells break down sugar with oxygen, transferring energy to ATP and releasing carbon dioxide, water, and heat. <i>Happens in plants, animals, and decomposers.</i>
Decomposition 1C	Food web 1C
The breakdown of dead material into simpler substances by decomposers. <i>A fallen log rots; its nutrients return to the soil.</i>	A diagram of feeding relationships. Each arrow points from the food to the eater. <i>Grass → rabbit → hawk.</i>

Trophic level 1C	ATP 1C
An organism's feeding position in a food chain. <i>Producer, primary consumer, secondary consumer.</i>	The molecule cells use as an immediate energy supply for work. <i>Muscle contraction uses ATP.</i>
Biodiversity 2A	Genetic diversity 2A
The variety of life: genes within a species, species within a community, and ecosystems within a region. <i>The three-scale picture.</i>	The variety of alleles within one species. <i>Rabbits of one species with different coat colors.</i>
Species diversity 2A	Ecosystem diversity 2A
The number of species in a community and how evenly individuals are spread among them. <i>Five pollinator species sharing visits evenly.</i>	The variety of ecosystems across a region. <i>Forest, wetland, and grassland in one county.</i>
Resilience 2A	Ecosystem services 2A
The ability of an ecosystem to keep or recover its functions after a disturbance. <i>Pollination continues after drought because several pollinator species respond differently.</i>	Benefits people get from ecosystems: provisioning, regulating, supporting, and cultural. <i>Food, clean water, crop pollination, and recreation.</i>

Pollination 2A	Keystone species 2A
Transfer of pollen to the receptive part of a flower, allowing fertilization and seed formation. <i>A bee carrying pollen between blossoms.</i>	A species whose effect on the community is large compared with its abundance. <i>A predator that keeps herbivores from overgrazing.</i>
Foundation species 2A	CHIPPO 2B
A species that creates or defines the habitat that many others depend on. <i>Reef-building corals.</i>	A memory aid for six threats: Climate change, Habitat loss, Invasive species, Pollution, Population, Overexploitation. <i>The table gives a response for each.</i>
Habitat loss 2B	Habitat fragmentation 2B
Removal or degradation of the places organisms live. <i>A forest cleared for farmland.</i>	Division of habitat into separated patches, which limits movement and gene flow. <i>A highway splitting a forest.</i>
Invasive species 2B	Overexploitation 2B
An introduced species that spreads and harms native species or ecosystems. <i>Not every introduced species is invasive; the harm matters.</i>	Harvesting a population faster than it can replace its losses. <i>Overfishing a breeding population.</i>

 Overconsumption 2B	 Endangered 2B
Using resources beyond what is needed or can be sustained. <i>Per-person use matters, not just the number of people.</i>	A species at high risk of extinction. <i>A small, isolated population losing habitat.</i>
 Extinct 2B	 Wildlife corridor 2B
No living individuals of the species remain. <i>Extinction cannot be reversed by conservation.</i>	A strip of habitat or a crossing that connects separated patches so organisms can move between them. <i>The Banff overpass.</i>
 Claim, evidence, reasoning (CER) 2B	 Population 2C
A claim answers the question; evidence is the data; reasoning explains how a scientific principle links them. <i>The flower-strip example.</i>	All the individuals of one species living in a defined area. <i>The rabbits in one meadow.</i>
 Immigration / emigration 2C	 Exponential growth 2C
Immigration: individuals moving into a population. Emigration: individuals moving out. $\Delta N = (B + I) - (D + E)$	Growth by a constant percentage, giving a J-shaped curve; bigger populations add more individuals each interval. <i>Bacteria doubling in fresh broth.</i>

 Logistic growth	2C  Limiting factor
Growth that slows as resources become limiting, giving an S-shaped curve that levels off near carrying capacity. <i>The idealized model, not an exact trace of real populations.</i>	Anything that restricts population growth: food, water, space, disease, or weather. <i>Food supply after a drought.</i>
2C  Density-dependent / independent	2C  Carrying capacity (K)
Dependent: the effect per individual changes with crowding. Independent: the effect is not caused by crowding. <i>Dependent: competition, disease. Independent: freeze, flood.</i>	The population size an environment can support over time with its resources and conditions. It can change. <i>K falls when habitat is lost.</i>
2C  Ecological footprint	2C  Biocapacity
An estimate of the productive land and water needed to supply what a person or population uses and to absorb its wastes. <i>Compared with biocapacity.</i>	The ability of ecosystems to regenerate resources and absorb wastes in a year. <i>Regrowth of forests and crops; absorption of carbon dioxide.</i>
2C  Earth Overshoot Day	2C  Environmental justice
The estimated date when humanity's demand for the year exceeds what Earth can regenerate that year. <i>An earlier date means greater demand relative to regeneration.</i>	Fair treatment and meaningful participation of all people in environmental decisions, including who bears the burdens and who receives the benefits. <i>One community drinks the polluted water while many use the factory's products.</i>

Academic  Assess; assessment Evaluate understanding or performance against criteria.	Academic  Commitment; effective Sustained effort; successful in achieving an intended result.
Academic  Condition; criteria A circumstance affecting a system; standards used to judge a claim or classify something.	Academic  Stable; constant; range Stable permits bounded variation; constant is unchanging; range is the span of values.
Academic  Dynamic; equilibrium Dynamic means changing or involving ongoing processes. Equilibrium describes opposing processes with no net change; homeostasis generally requires ongoing regulation.	Academic  Mechanism; model A process explaining how an effect occurs; a simplified representation used to explain or predict.
Academic  Reinforce Strengthen a response or change, as in positive feedback.	Academic  Affect; effect Affect is usually a verb meaning influence; effect is usually a noun meaning result.

Academic  Convert; transfer; store Change form; move between locations or organisms; retain for later use. Energy is transformed or transferred, not created by organisms.	Academic  Obtain; release; generate Acquire; let out; produce. Say organisms generate ATP or heat, not that they create energy from nothing.
Academic  Cycle; process; interact Move through repeated transfers; a sequence of events; influence one another.	Academic  Efficient Producing a given output with less input or waste; specify which input and output.
Academic  Complex; context; diverse Having many interacting parts; relevant circumstances; varied in composition.	Academic  Collapse; resilient Severe loss of structure or function; able to retain or recover function after disturbance.
Academic  Resources Materials, energy sources, or conditions organisms use to survive and reproduce.	Academic  Consumption; exploitation Use of resources; use or harvest of organisms or resources. Overexploitation exceeds replacement capacity.

Academic  Destruction; disruption Removal or severe damage; interference with a process or relationship.	Academic  Impact; threat An effect; a potential cause of harm. State the affected organism or process.
Academic  Migration Movement between locations, often seasonal; immigration and emigration specify movement relative to a defined population.	Academic  Prevention; restoration Avoiding a problem; helping a damaged ecosystem recover structure or functions.
Academic  Capacity; finite The amount that can be supported or held; limited in quantity.	Academic  Decline; fluctuate Decrease; vary upward and downward over time.
Academic  Exacerbate; exceed Make a problem worse; go beyond a limit.	Academic  Factor; outcome Something contributing to a result; the result itself.

Academic  Indefinitely; limit Without an end; a constraint or boundary. Exponential growth cannot continue indefinitely with finite resources.	Academic  Replenish Replace resources as they are used, through regeneration or other supply.
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